



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2010
Year 11 Yearly
Examination

Mathematics Extension Continuers

(3 Unit)

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question.
- All answers to be given in simplified exact form unless otherwise stated.
- Hand in your answers in 3 separate bundles:
Section A (Question 1 and Question 2),
Section B (Question 3 and Question 4) and
Section C (Question 5 and Question 6)

Total Marks – 82

- Attempt questions 1-6
- All questions are **NOT** of equal value.

Examiner: *P. Bigelow*

Section A – Start a new booklet.

Question 1 (14 marks).	Marks
a) Solve $x^2 + 2x - 8 = 0$.	1
b) Find T_{10} of $5 + 9 + 13 + 17 + \dots$	1
c) If $f(y) = 9 - y^2$, find: (i) $f(-2)$ (ii) $f(y+1)$	2
d) State the domain of $f(x) = \sqrt{3-x}$.	1
e) Write down the equation of the parabola with focus (0,5) and directrix $y + 5 = 0$.	1
f) Find x in the following: (i) $\log_x 36 = 2$ (ii) $\log_8 128 = x$	2
g) Find the exact value of $\tan 15^\circ$.	2
h) Solve: (i) $ 3x-1 > 7$ (ii) $\frac{4}{x} \leq \frac{3}{x+1}$	4

End of Question 1

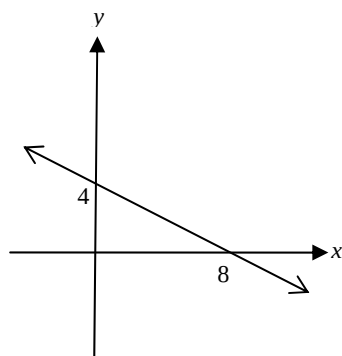
Question 2 (12 Marks).**Marks**

a) Find the sum of the first 9 terms of $2 - 1 + \frac{1}{2} - \frac{1}{4} + \dots$

1

b) Write down the equation of the line represented in the diagram:

2



c) State whether the following functions are ODD, EVEN or NEITHER:

3

(i) $f(x) = \frac{4}{9 + x^2}$

(ii) $f(x) = \frac{x}{9 + x}$

(iii) $f(x) = \frac{-x}{9 + x^2}$

d) Solve $2^{3x+2} = 64$.

2

e) By considering $0.\dot{5}\dot{4}$ as an infinite geometric series, express $0.\dot{5}\dot{4}$ as a fraction in simplest form.

2

f) Simplify:

2

(i) $\sin 5A \cos 2A - \cos 5A \sin 2A$

(ii) $2 \sin 3\theta \cos 3\theta$

End of Question 2**End of Section A**

Section B – Start a new booklet.

Question 3 (11 marks).	Marks
a) Sketch the following on separate diagrams (showing essential features):	4
(i) $xy = -4$	
(ii) $y = 2^{-x}$	
(iii) $y = \sqrt{4 - x^2}$	
(iv) $x^2 + y^2 - 6y = 0$	
b) State whether the following quadratics are INDEFINITE, POSITIVE DEFINITE or NEGATIVE DEFINITE:	3
(i) $2x^2 + 3x + 7$	
(ii) $6 - x - x^2$	
(iii) $x^2 - 9x - 8$	
c) Solve: $4^x - 9(2)^x + 8 = 0$.	2
d) An interval PQ is divided externally in the ratio 4:3 by the point S . Find S , if P is $(4,3)$ and Q is $(-1,9)$.	2

End of Question 3

Question 4 (16 marks).**Marks**

- a) Graph the region defined by the intersection of $y \geq x^2$ and $x + y \leq 2$. 2
- b) Find the acute angle between the lines: $2x - y - 5 = 0$ and $x - 3y + 3 = 0$. 2
- c) Find $f'(x)$ in the following: 6
- (i) $f(x) = 3x^2 - x + 1$
- (ii) $f(x) = (1 - 5x)^6$
- (iii) $f(x) = x\sqrt{1+x}$
- (iv) $f(x) = \frac{3x-1}{3x+2}$
- d) If $\sin x = \frac{3}{4}$ and $\frac{\pi}{2} \leq x \leq \pi$, find the exact value of $\sin 2x$. 2
- e) If α and β are the roots of $2x^2 - 6x - 1 = 0$, find: 4
- (i) $\alpha + \beta$
- (ii) $\alpha\beta$
- (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$
- (iv) $\alpha^2 + \beta^2$

End of Question 4**End of Section B**

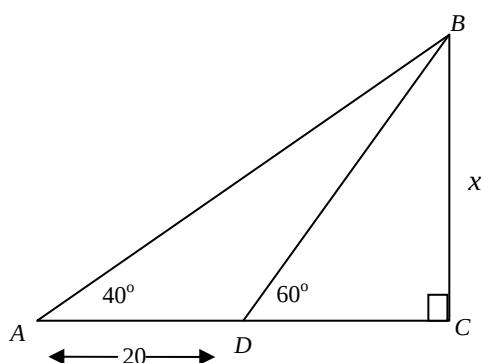
Section C – Start a new booklet.

Question 5 (15 marks).

Marks

- a) Find the value of x correct to 3 significant figures.

3



- b) Sketch $(x-2)^2 = 8(y+1)$, showing vertex, focus and directrix.

2

- c) (i) Express $\sin \theta + \cos \theta$ in the form $R \sin(\theta + \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

4

- (ii) Hence, solve $\sin \theta + \cos \theta = 1$ for $0 < \theta < 2\pi$.

- d) Find the general solution of $\tan x = 1$

2

- e) (i) Find the axis of symmetry of $y = 4 + x - x^2$

2

- (ii) Hence, or otherwise, find the maximum value of $4 + x - x^2$

2

End of Question 5

Question 6 (14 marks).**Marks****a)** Prove the following:

4

(i) $\cos^4 x - \sin^4 x = \cos 2x$

(ii) $\frac{\sin 2A}{1 - \cos 2A} = \cot A$

b) Given $5^x = 13$, find x correct to two decimal places.

2

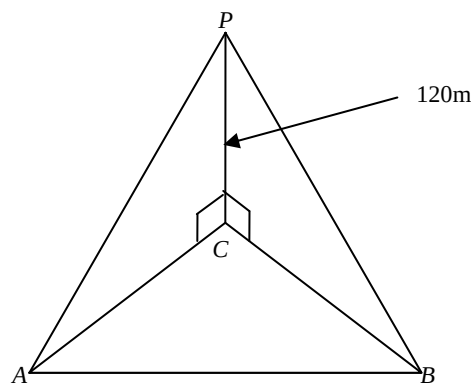
c) A couple wish to save for a deposit on a home. They need to save \$20,000 over a 5 year period. They deposit \$ P , every month, into an account which is paying 9%p.a., compounding monthly.

4

(i) Show that $20000 = P(1.0075 + 1.0075^2 + \dots + 1.0075^{60})$.

(ii) Find P to the nearest dollar.**d)** Two boats at A and B are observed from the top P of a vertical cliff CP of height 120 metres. A is on a bearing of 195°T from the cliff and its angle of depression from P is 22° . B is on a bearing of 161°T from the cliff and its angle of depression from P is 27° .

4

(i) Find $\angle ACB$.

(ii) Use the cosine rule to find the distance between the boats (to the nearest metre).

End of Question 6.**End of Section C.****End of Examination.**

Mathematics Extension Continuers 2010 – Section A:

Question 1:

a) $x^2 + 2x - 8 = 0$

$$(x + 4)(x - 2) = 0$$

$$x = -4, \quad 2$$

b) $a = 5, d = 4$

$$T_{10} = 5 + 9 \times 4$$

$$T_{10} = 41$$

c) $f(y) = 9 - y^2$

(i) $f(-2) = 9 - (-2)^2$

$$f(-2) = 5$$

(ii) $f(y + 1) = 9 - (y + 1)^2$
 $= 9 - (y^2 + 2y + 1)$
 $= 9 - y^2 - 2y - 1$
 $= 8 - 2y - y^2$

d) $f(x) = \sqrt{3 - x}$

$$\text{Domain: } x \leq 3$$

e) $x^2 = 20y$

f)

(i) $\log_x 36 = 2$

$$36 = x^2$$

$$x = 6$$

(ii) $\log_8 128 = x$

$$\log_8 8^{7/3} = x$$

$$\frac{7}{3} \log_8 8 = x$$

$$x = \frac{7}{3}$$

g) $\tan 15^\circ = \tan(45^\circ - 30^\circ)$

$$= \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$$

$$= \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \times \frac{1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$= 2 - \sqrt{3}$$

Question 2:

a) $2 - 1 + \frac{1}{2} - \frac{1}{4} + \dots$

$$a = 2$$

$$r = -\frac{1}{2}$$

$$S_9 = \frac{2 \left(1 - \left(-\frac{1}{2} \right)^9 \right)}{1 + \frac{1}{2}}$$

$$S_9 = \frac{171}{128}$$

b) $m = -\frac{1}{2}$

$$y - 0 = -\frac{1}{2}(x - 8)$$

$$2y = -x + 8$$

$$x + 2y - 8 = 0$$

c)

(i) $f(x) = \frac{4}{9+x^2}$

$$\begin{aligned} f(-x) &= \frac{4}{9+(-x)^2} \\ &= \frac{4}{9+x^2} \end{aligned}$$

$$\text{Since } f(x) = f(-x)$$

$\therefore f(x)$ is even

(ii) $f(x) = \frac{x}{9+x}$

$$f(-x) = \frac{-x}{9-x}$$

$$-f(x) = -\frac{x}{9+x}$$

$$\text{Since } f(x) \neq f(-x) \neq -f(x)$$

$\therefore f(x)$ is neither.

(iii) $f(x) = \frac{-x}{9+x^2}$

$$\begin{aligned} f(-x) &= \frac{-(-x)}{9+(-x)^2} \\ &= \frac{x}{9+x^2} \end{aligned}$$

$$\begin{aligned} -f(x) &= -\frac{-x}{9+x^2} \\ &= \frac{x}{9+x^2} \end{aligned}$$

$$\text{Since } f(-x) = -f(x)$$

$\therefore f(x)$ is odd.

d) $2^{3x+2} = 64$

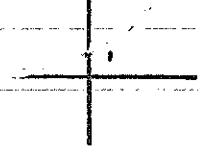
$$2^{3x+2} = 2^6$$

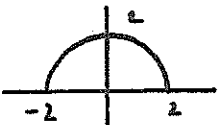
$$3x + 2 = 6$$

$$x = \frac{4}{3}$$

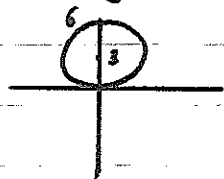
QUESTION THREE

i)  $xy = -4$

ii)  $y = 2^{-x}$

iii)  $y = \sqrt{4-x^2}$

iv) $x^2 + y^2 - 6y = 0$
 $x^2 + y^2 - 6y + 9 = 9$
 $x^2 + (y-3)^2 = 9$



b) $y = 2x^2 + 3x + 7$, $\Delta = -47$
 i) Positive Definite

ii) $y = -x^2 - x + 6$, $\Delta = 25$
 Indefinite

iii) $y = x^2 - 9x - 8$, $\Delta = 113$
 Indefinite

c) $4^x - 9(2^x) + 8 = 0$
 $(2^x)^2 - 9(2^x) + 8 = 0$
 let $u = 2^x$

$$u^2 - 9u + 8 = 0$$

$$(u-8)(u-1) = 0$$

$$u = 8, 1$$

$$x = 3, 0$$

d) P(4, 3) Q(-1, 9)

$$m:n = 4:-3$$

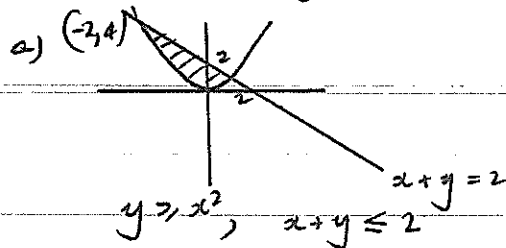
$$x = \frac{-3 \times 4 + 4 \times -1}{1}, y = \frac{-3 \times 3 + 4 \times 9}{1}$$

$$= -16$$

$$= 27$$

$$(x, y) = (-16, 27)$$

QUESTION FOUR



b) $y = 2x - 5$ $\tan A = 2$
 $y = \frac{1}{3}x + 1$ $\tan B = \frac{1}{3}$
 $\tan(A-B) = \frac{2 - \frac{1}{3}}{1 + \frac{2}{3}} = 1$

$$A - B = 45^\circ = \frac{\pi}{4}$$

c) i) $y = 3x^2 - x + 1$ $y' = 6x - 1$

ii) $y = (1-5x)^6$ $y' = -30(1-5x)^5$

iii) $y = x(1+x)^{1/2}$, $y' = x \cdot \frac{1}{2}(1+x)^{-1/2} + (1+x)^{1/2}$
 $y' = \frac{x}{2\sqrt{1+x}} + \sqrt{1+x}$

$$= \frac{x + 2(1+x)}{2\sqrt{1+x}} = \frac{3x+2}{2\sqrt{1+x}}$$

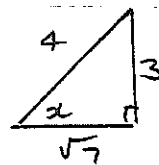
iv) $y = \frac{3x-1}{3x+2}$, $y' = \frac{(3x+2)3 - 3(3x-1)}{(3x+2)^2}$
 $= \frac{9}{(3x+2)^2}$

d) $\sin x = \frac{3}{4}$ $\frac{\pi}{2} \leq x \leq \pi$
 $x \approx 131^\circ$, $2x \approx 263^\circ$ Hence $\sin 2x < 0$

$$\sin 2x = 2 \sin x \cos x$$

$$= 2 \times \frac{3}{4} \times \frac{\sqrt{7}}{4}$$

$$= -\frac{3\sqrt{7}}{8}$$



e) $2x^2 - 6x - 1 = 0$, $\alpha + \beta = 3$, $\alpha\beta = -\frac{1}{2}$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{3}{-\frac{1}{2}} = -6$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 9 - 2(-\frac{1}{2})$$

$$= 10$$

SECTION C

Q5

$$(a) \tan 60^\circ = \frac{x}{DC}$$

$$DC = \frac{x}{\tan 60^\circ}$$

$$\tan 40^\circ = \frac{x}{20+DC}$$

$$\text{So } (20+DC) = \frac{x}{\tan 40^\circ}$$

$$20 + \frac{x}{\tan 60^\circ} = \frac{x}{\tan 40^\circ}$$

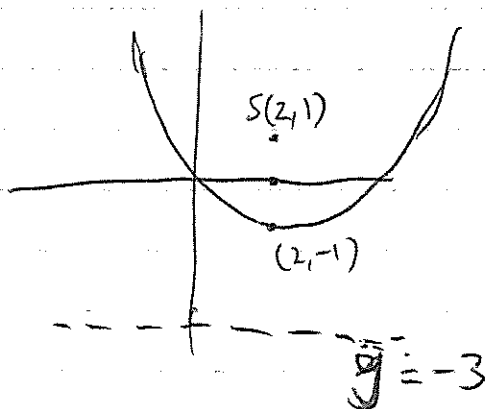
$$20 = x \left(\frac{1}{\tan 40^\circ} - \frac{1}{\tan 60^\circ} \right)$$

$$x = \frac{20}{\frac{1}{\tan 40^\circ} - \frac{1}{\tan 60^\circ}}$$

$$= \frac{20(\tan 40^\circ \tan 60^\circ)}{\tan 60^\circ - \tan 40^\circ}$$

$$= 32.6$$

(b)



$$(c)(i) R \sin(\theta + \alpha) = R \sin \theta \cos \alpha + R \sin \alpha \cos \theta$$

$$R \sin \alpha = 1$$

$$R \cos \alpha = 1$$

$$\tan \alpha = 1.$$

$$\alpha = 45^\circ.$$

$$R^2 = 2$$

$$R = \sqrt{2}.$$

$$\sqrt{2} \sin(\theta + 45^\circ).$$

$$(ii) \sqrt{2} \sin(\theta + 45^\circ) = 1$$

$$\sin(\theta + 45^\circ) = \frac{1}{\sqrt{2}}.$$

$$\theta + 45^\circ = 45^\circ,$$

$$\theta + \frac{\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4},$$

$$\theta = \frac{\pi}{2}.$$

$$(d) \tan x = 1$$

$$x = \frac{\pi}{4} + \pi n \quad n \in \mathbb{Z}$$

$$(e) (i) y = 4 + x - x^2.$$

$$x = \frac{-b}{2a}.$$

$$x = \frac{-1}{-2}.$$

$$x = \frac{1}{2}.$$

$$(ii) 4 + \frac{1}{2} - \frac{1}{4}.$$

$$= 4\frac{3}{4}.$$

$$Q6 (a) (i) \cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x).$$

$$= \cos^2 x - \sin^2 x$$

$$= \cos 2x.$$

$$(ii) \frac{\sin 2A}{1 - \cos 2A} = \frac{2 \sin A \cos A}{\cos^2 A + \sin^2 A - \cos^2 A + \sin^2 A}.$$

$$= \frac{2 \sin A \cos A}{2 \sin^2 A}$$

$$= \frac{\cos A}{\sin A}$$

$$= \cot A.$$

$$(b) 5^x = 13.$$

$$x = \frac{\log 13}{\log 5}.$$

$$= 1.60.$$

$$(c) (i) A_1 = P \times 1.0075.$$

$$A_2 = (A_1 + P) 1.0075.$$

$$= P \times 1.0075^2 + P \times 1.0075$$

$$A_3 = P \times 1.0075^3 + P \times 1.0075^2 + P \times 1.0075.$$

$$A_{60} = P(1.0075^{60} + 1.0075^{59} + \dots + 1.0075).$$

$$\text{Since } A_{60} = 20000.$$

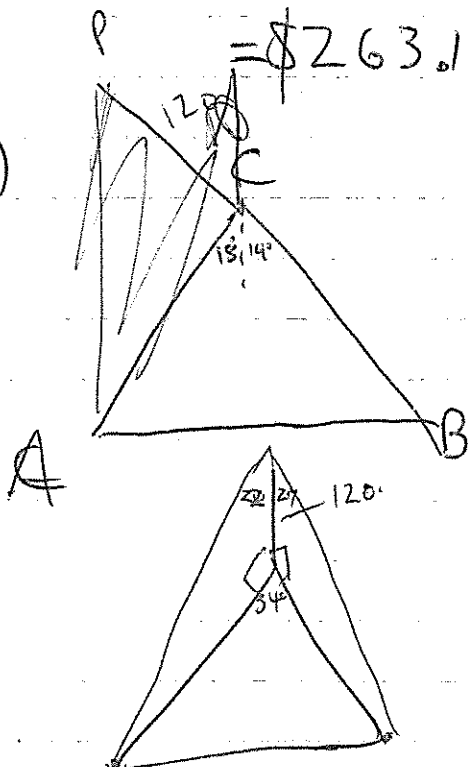
$$\text{Then } 20000 = P(1.0075 + \dots + 1.0075^{60}).$$

$$(ii) \quad 20000 = P \left(\frac{1.0075(1 - (1.0075)^{60})}{1 - 1.0075} \right)$$

$$P = \frac{20000(-0.0075)}{1.0075(1 - 1.0075^{60})}.$$

$$P = \$263.19.$$

(d)



$$(i) 34^\circ.$$

$$(ii), \tan 22 = \frac{AC}{120} \Rightarrow AC = 120 \tan 22$$

$$\tan 27 = \frac{BC}{120} \Rightarrow BC = 120 \tan 27.$$

$$AB^2 = AC^2 + BC^2 - 2 \times AC \times BC \cos 34.$$

$$AB = 34.26.$$